

and permanent embedding within the equipment of interest.

Scheduling of measurements

Another limitation of the handheld probe approach is the lack of real-time notification of troublesome vibration shifts. The same is true for most piezo-based sensors, which are typically at a very low level of integration (transducer only in some cases) with the data transferred elsewhere for later analysis. These devices require external intervention and thus present an opportunity for missed events/shifts.

embedded FFT analysis at the sensor, vibration shifts can be isolated to specific sources immediately. Such a fully-integrated sensor element could also reduce development time for equipment designers by six to twelve months, given the completeness and simplicity of a fully-integrated and autonomous sensor.

Accessing the data

Embedded FFT analysis assumes that the analogue sensor data has been conditioned and converted to digital, and thus data transmission is greatly simplified. In fact most vibra-

Use of vibration sensors can reduce avoidable equipment repairs, which are a major contributor to lost productivity in industrial manufacturing processes

On the other hand, an autonomous sensor processing system that includes sensor, analysis, storage and alarm capability, and is still small enough for embedding, offers the fastest notification of vibration shifts, as well as the best ability to show time-based trends.

Understanding the data

The ideal real-time notification from an embedded sensor is only possible if frequency domain analysis is employed. Any given equipment typically has multiple sources of vibration such as bearing defect, imbalance, gear mesh and by-design sources such as a drill or machine press producing vibration in normal course of operation. A time-based analysis of the equipment produces a complex waveform, combining these multiple sources, which provides little discernible information prior to fast fourier transform (FFT) analysis.

Most piezo-based sensor solutions rely on external computation and analysis of the FFT. This not only eliminates the possibility of real-time notification but also puts substantial additional design burden on the equipment developer. With

tion sensor solutions in use today are analogue output only leading to signal degradation during transmission. Not to mention the complexity of off-line data analysis.

Most industrial equipment that require vibration monitoring tend to exist in noisy, moving, inaccessible and even dangerous environments. Thus, there is a strong desire to not only reduce the complexity of interface cabling but also perform as much of the data analysis as possible at-the-source to capture the most accurate representation of the equipment vibration as possible.

How much data

Many existing sensor solutions are single-axis piezo transducers. These piezo sensors provide no directionality information and thus limit the understanding of the equipment vibration profile. The lack of directionality translates to the need for very low-noise sensors, which also impact cost. The availability of tri-axis microelectromechanical systems (MEMS)-based sensors allow a significant increase in the ability to isolate the vibration source, while potentially improving cost.

Probe points

The question of where to place the sensors is critical but highly dependent on the type of equipment, environment and even the life cycle of the equipment. With existing high-cost sensor elements limiting the number of probe points to a few or one, placement is more critical. This translates to either significant additional up-front development time to determine optimal placement through experimentation or in most cases leads to some compromise in the amount and quality of the data to be captured.

The existence of more fully-integrated sensor-probes at a fraction of existing costs can allow placement of multiple probes per system. Also, it can lessen up front development time/cost or simply lead to fewer and less costly sensors.

Equipment life-cycle shifts

The transducer element, regardless of technology, is an important consideration. But typically more critical is the sensor signal-conditioning and processing wrapped around the transducer. The signal conditioning and processing is not only specific to the unique equipment but also to the life-cycle of the equipment. This translates to several important considerations in the design of the sensor.

Earlier analogue-to-digital conversion (i.e., at the sensor head, versus off-equipment) allowed configuration/tuning in-system. An ideal sensor would provide a simple programmable interface, which would simplify the equipment set-up through quick baseline data captures, manipulation of filtering, programming of alarms and experimentation with different sensor locations.

The existing simple sensors are configurable at equipment set-up to an extent. However, some compromise in sensor settings must still be made to accommodate changes in maintenance concerns over the life of the equipment. For example, should the sensor be configured for early life